A CONDITIONAL LOCALIZED ENTROPY FUNCTIONAL AND STATE-SPACE COMPACTIFICATION FOR NAVIER–STOKES SINGULAR BRANCHES

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Abstract. Assuming the companion Navier–Stokes manuscripts on local Type I reduction [4], residual Type I exclusion [3], and Type II regularity [5], we construct a localized entropy functional $\mathcal{W}$ on the singular-branch state space supplied by those works. The functional combines local enstrophy, a localized critical-norm velocity budget in L^3, mean-free local pressure in $L^{3/2}$, and a normalized backward adjoint weight adapted to the modulated drift of the repaired-gauge representation. Under the imported companion estimates, the time derivative of $\mathcal{W}$ decomposes into an explicit nonnegative dissipation, a cutoff-normalizer term controlled by the local Caccioppoli estimate, a harmonic-pressure remainder controlled by the localized pressure decomposition, a modulation correction controlled by the moving corrected monotonicity, and a Type I rescaling defect controlled by the velocity no-escape estimate. We then identify, in the constrained first variation of $\mathcal{W}$, three Euler–Lagrange residuals – velocity, pressure, and adjoint – whose simultaneous vanishing characterizes a quantitative *zero-gradient balance*. This balance is matched row-by-row with the five terminal model classes of the companion architecture, and entropy-critical limits are assigned to the first applicable class in the order (1)–(5). The localized entropy is therefore a conditional organizing quantity for the companion program, not a replacement for its local PDE estimates.

Contents

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1. Introduction

The local study of possible finite-time singularities for the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \qquad \operatorname{div} u = 0, \tag{1.1}$$

on $\mathbb{R}^3 \times (T - \delta, T)$ is governed by a persistent tension between scale-critical compactness and the nonlocal structure of the pressure. Since Leray's construction of global weak solutions [17], the basic local framework has been provided by suitable weak solutions and the partial regularity results of Scheffer, Caffarelli–Kohn–Nirenberg, Lin, and subsequent authors [21, 7, 18, 11]. In that framework, the relevant local quantities are the scale-invariant velocity and pressure functionals, the local energy inequality, and the compactness mechanisms surviving parabolic rescaling. The pressure term is indispensable but nonlocal, and that is precisely what complicates a clean compactification of singular branches.

The companion manuscripts [4, 3, 5] respond to this difficulty by a detailed local state-space stratification. The setup paper [4] reduces a singular branch either to a bounded centered ancient profile produced by Seregin-type extraction, or to a local Type II branch. The residual paper [3] closes the remaining Type I residual strata. The Type II paper [5] proves the repaired-gauge representation, the localized pressure decomposition, the compact-cylinder Caccioppoli estimate, the multibubble and cascade exclusions, the scale-collapse and scale-rigid routing, the windowwise carrier package, the corrected monotonicity, and the terminal critical-profile decomposition. Together these papers yield a finite decision tree.

Purpose of this paper. The companion papers organize the singularity problem as a decision tree of named local strata. Conditional on the estimates and closure theorems imported from those papers, this paper shows that the same decision tree can be read off from a *single monotone scalar quantity* – a localized entropy $\mathcal{W}$. The point is not to bypass the local PDE estimates but to package them as the error budget of one identity. A reader who accepts the companion estimates obtains, from the present paper, a one-quantity description of the entire stratification.

The motivation comes from two classical sources. Following Arnold's formal interpretation of Euler flow as geodesic motion on the group of volume-preserving diffeomorphisms [2, 9], one may regard the convective term as a transport term built into the ambient geometry. Following Hamilton and Perelman [12, 20], one expects a useful organizing quantity to couple a backward adjoint weight to the dissipative part of the equation and to penalize failure of compactification. In the present setting, however, the functional must be strictly local and compatible with the actual PDE estimates available in the companion papers. We therefore work with local enstrophy, a localized critical-norm velocity budget, and mean-free local pressure rather than with a global geometric invariant, and we measure the adjoint concentration via a Fisher-information component associated with the local backward adjoint density on a compact cylinder.

What is rigorously proved here. The paper is self-contained at the level of the entropy formalism, while its local PDE inputs are the imported companion packages stated in Theorems 2.1 to 2.4. Specifically, we prove:

(1) the f-equation associated with the normalized backward adjoint weight (Theorem 3.4);
(2) a localized monotonicity identity for $\mathcal{W}$ with explicit error decomposition (Theorem 4.6), with each error term tagged to a specific companion theorem;
(3) the exact constrained first variation (Theorem 5.1) in a fixed independent-variable framework;
(4) a conditional matching of the zero-gradient balance with the five terminal model classes (Theorem 5.9), with the matching specified by an explicit topology and table (Table 1);
(5) an ordered-classification theorem assigning every entropy-critical limit to the first applicable terminal class (Theorem 6.6).

The local PDE estimates underlying these statements live in the companion papers; Section 2 restates the exact inputs used here and cites them by theorem number.

Main result.

Theorem 1.1 (Conditional entropy reduction of the companion program)**.** *Within the singular-branch framework of [4, 3, 5], there is a localized entropy functional $\mathcal{W} = \mathcal{W}_{\chi,R}[V, P, \rho](s; \sigma)$ with the following properties.*

(i) ***Monotonicity (Theorem 4.6).*** *Along every admissible branch state,*

$$\mathcal{W}(s_1; \sigma) - \mathcal{W}(s_2; \sigma) \geq \int_{s_1}^{s_2} \mathcal{D}_{\chi,R}(\tau; \sigma)\, d\tau - \mathcal{E}_{\chi,R}(s_1, s_2; \sigma),$$

with $\mathcal{D}_{\chi,R} \geq 0$ explicit and $\mathcal{E}_{\chi,R} = \mathcal{E}^{\mathrm{cut}} + \mathcal{E}^{\mathrm{prs}} + \mathcal{E}^{\mathrm{mod}} + \mathcal{E}^{\mathrm{T1}}$ a sum of four explicit terms, each controlled by a named companion theorem.

(ii) ***First variation (Theorem 5.1).*** *The constrained first variation of $\mathcal{W}$ decomposes into three displayed Euler–Lagrange residuals $\mathfrak{G}_{\chi,R} = \big(\mathfrak{M}^{\mathrm{vel}}, \mathfrak{M}^{\mathrm{prs}}, \mathfrak{M}^{\mathrm{adj}}\big)$, written out in the fixed independent-variable framework in (5.5)–(5.7).*

(iii) ***Compactification (Theorem 6.5).*** *Every admissible singular branch with finite corrected entropy and error budget has an entropy-critical subsequential limit.*

(iv) ***Conditional classification (Theorems 5.9 and 6.6).*** *Using the imported row-entry dictionary, an admissible zero-gradient branch state lies in one of five terminal model classes: (1) compact single-core, (2) multiple-core, (3) rotational or stationary-hull, (4) scale-cascade, (5) diffuse or tail-defect. The correspondence between vanishing components of $\big(\mathcal{E}_{\chi,R}, \mathfrak{M}^{\mathrm{vel}}, \mathfrak{M}^{\mathrm{prs}}, \mathfrak{M}^{\mathrm{adj}}\big)$ and the five classes is recorded in Table 1.*

(v) ***Ordering (Theorem 6.6).*** *Every entropy-critical limit lies in the* first *applicable class in the order (1)–(5).*

(vi) ***Closure (Theorem 7.1).*** *Each of the five terminal model classes is excluded, under the imported hypotheses, by the terminal-class exclusion package Theorem 2.5.*

Remark 1.2. Theorem 1.1 is an organizational theorem: the local PDE content remains in the companion papers. What is new here is the entropy formulation, the explicit error decomposition, and the explicit correspondence between Euler–Lagrange residuals and terminal model classes.

Relation to the literature. The local Navier–Stokes background used throughout consists of Leray's weak solutions [17], the suitable weak solution framework and partial regularity results [21, 7, 18], classical critical-space and local theory [15, 24, 8, 13, 16], the endpoint regularity theorem of Escauriaza–Seregin–Šverák [10], the self-similar rigidity theorem of Nečas–Růžička–Šverák [19], the Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [14], and the local Type I and ancient-solution inputs of Albritton–Barker and Seregin [1, 22]. The pressure terms in the present functional are measured in the same mean-free local norms controlled by the Calderón–Zygmund estimates [6, 23] already used in [5].

Structure of the paper. Section 2 records the companion results imported into the entropy formulation, with theorem-numbered package statements. Section 3 defines the local adjoint weight, derives the f-equation, and defines the localized entropy functional. Section 4 carries out the explicit $\frac{d}{ds}\mathcal{W}$ calculation, displays the four-piece error decomposition, and proves the localized monotonicity identity. Section 5 computes the constrained first variation and displays the three Euler–Lagrange residuals in the fixed independent-variable framework. Section 6 extracts entropy-critical limits, identifies them with the five terminal model classes via an explicit correspondence table, and proves the ordered classification. Section 7 applies the imported terminal-class exclusion package to exclude each terminal class under the stated hypotheses.

2. Companion Input and the Admissible State Space

We import companion results as named theorem packages. Each package below restates only the hypotheses and conclusions used in the entropy argument and cites the stable theorem numbers in the companion manuscripts. Later sections refer to these imported packages by the labels introduced here, rather than repeating long source-paper citations.

Theorem 2.1 (Imported setup and branch-reduction package)**.** *Let (u, p) be a suitable weak Navier–Stokes solution with a hypothetical singular point. The setup paper supplies the scale-invariant local boundedness criteria, the positive concentration token at a singular point, the local Type I/Type II branch dichotomy, the local Seregin extraction in the Type I branch, the pointwise Type I velocity no-escape estimate, and the direct Liouville closures for the small, stationary, uniformly tight, and structure-controlled bounded centered ancient Type I profiles. These are* [4, Theorems 4.1, 4.2, 5.7, 6.5, 7.3, 8.4; Lemma 8.7; Theorems 15.7–15.9, 21.4, and 25.5].

Theorem 2.2 (Imported represented Type II analytic package)**.** *Every local Type II branch produced by Theorem 2.1 admits the repaired-gauge representation used in this paper, the local pressure decomposition and pressure-normalization comparison, the stability topology for subsequences, the renormalized local energy inequality, the compact-cylinder Caccioppoli estimate, the fixed- and moving-cutoff corrected monotonicity formulae, and the ordered local state-space decomposition. These are* [5, Theorems 3.26, 3.36–3.38, 5.4, 9.11, 9.19, 10.3, 10.4, 10.17, 10.18, and 10.25; Propositions 3.35 and 5.1; Lemma 5.3].

Theorem 2.3 (Imported Type II terminal-closure package)**.** *In the represented Type II state space of Theorem 2.2, the compact single-core, multibubble, same-point cascade, separated-point, rough-core, scale-collapse, scale-rigid, windowwise carrier, cost-divergence, critical-profile, retained compact-test, diffuse-tail, exterior, and final local Type II terminal alternatives are either routed to a previous alternative or excluded. This is the combined content of* [5, Theorems 2.19, 4.19, 4.20, 4.45, 5.4, 5.7, 6.7, 6.20, 6.24, 7.130, 8.6, 8.14, 8.38, 8.41, 8.46, 9.42, 9.82, 9.85, 9.89, 9.91, 9.94, 9.103, 9.113, 9.134, 9.139, 10.5, 10.6, 10.38, and 10.39; Corollaries 7.123, 7.128, 9.43, and 9.142; Proposition 4.43].

Theorem 2.4 (Imported residual Type I closure package)**.** *For a residual Type I branch satisfying the setup normalization and retained local L^3-mass hypotheses, the residual state-space stratification closes the rotational, stationary-hull, affine/parasitic, critical-tail, diffuse-defect, generic terminal, and remaining residual classes. In particular the residual hypothesis left open by Theorem 2.1 is verified. This is* [3, Theorems 1.14, 4.4, 9.27, 9.38, 9.91, 9.98, 9.113, 10.25, 11.2, 11.3, and 12.1; Propositions 6.10 and 7.1; Remark 7.4; Corollary 12.2]*.*

Theorem 2.5 (Imported terminal-class exclusion package)**.** *Under the admissibility, normalization, branch-topology, retained-mass, and finite-budget hypotheses used in this paper, the five terminal classes are excluded by the following companion results:*

Terminal class	*Companion manifestation*	*Exclusion input*
$\mathcal{C}_{\mathrm{core}}$	*bounded centered Type I profiles; compact single-core or retained compact-test Type II states*	[4, Theorems 15.7–15.9, 21.4, and 25.5]*;* [5, Theorems 2.19, 9.103, and 10.5]
$\mathcal{C}_{\mathrm{multi}}$	*multibubble, same-point cascade, separated-point, and terminal decoupling states*	[5, Theorems 4.19, 4.20, 4.45, 8.6, 8.14, and 10.6]
$\mathcal{C}_{\mathrm{rot}}$	*rotational, stationary-hull, and axisymmetric residual branches*	[3, Theorem 4.4; Propositions 6.10 and 7.1; Remark 7.4; Theorem 10.25]
$\mathcal{C}_{\mathrm{cascade}}$	*scale-collapse, scale-rigid, windowwise carrier, and cost-divergence states*	[5, Theorems 6.7, 6.20, 6.24, 7.130, and 9.42; Corollaries 7.123, 7.128, and 9.43; Proposition 4.43]
$\mathcal{C}_{\mathrm{tail}}$	*diffuse, critical-tail, exterior, and terminal inactive defects*	[3, Theorems 9.27, 9.38, 9.91, 9.98, and 9.113]*;* [5, Theorems 8.38, 8.41, 8.46, 9.89, 9.91, and 9.94]

Thus each of the terminal classes $\mathcal{C}_{\mathrm{core}}, \mathcal{C}_{\mathrm{multi}}, \mathcal{C}_{\mathrm{rot}}, \mathcal{C}_{\mathrm{cascade}}, \mathcal{C}_{\mathrm{tail}}$ has no admissible state satisfying these hypotheses.

Definition 2.6 (Admissible branch state)**.** An *admissible branch state* is either:

(i) a bounded centered ancient profile (V,P) on $\mathbb{R}^3 \times (-\infty, 0)$ produced by the Type I Seregin extraction imported in Theorem 2.1; in this case we set the modulation coefficients to the centered self-similar values $a(s) \equiv \frac{1}{2}$, $b(s) \equiv 0$, and set $\nu = 1$ in the unified equation below; or

(ii) a represented local Type II branch (V, P, a, b) on a compact parabolic cylinder $Q_R = B_R \times (-R^2, 0)$ in the sense of the represented Type II analytic package Theorem 2.2, satisfying the renormalized suitable equation

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P - \nu\Delta V = a(\tau)V + a(\tau)(y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \operatorname{div} V = 0, \tag{2.1}$$

with $\nu > 0$, $a, b \in L^\infty(-R^2, 0)$, the localized pressure decomposition, and the renormalized local energy inequality imported in Theorem 2.2.

We write $\mathfrak{S}$ for the collection of admissible branch states.

Remark 2.7. Setting $a \equiv \frac{1}{2}$, $b \equiv 0$, and $\nu = 1$, the renormalized equation (2.1) reduces to the centered self-similar equation

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P - \Delta V = \tfrac{1}{2}V + \tfrac{1}{2}(y\cdot\nabla V),$$

which is exactly the centered ancient equation produced by Seregin extraction. Thus the Type I and Type II cases are simultaneous specializations of (2.1), justifying the unified treatment below. All calculations of Sections 4 and 5 are performed on (2.1) and apply to both cases by setting (a, b, ν) to their respective values.

The entropy formalism does not introduce a new state space; it is defined on the state space already produced by Theorems 2.1 and 2.2. The role of the entropy is to organize which branch changes are dissipative, which are merely routing mechanisms, and which terminal regimes remain after all routing has been exhausted.

Remark 2.8 (Regularity convention). The differential identities in Sections 3 to 5 are first stated and proved for smooth represented branch states and smooth positive adjoint weights on compact subcylinders. For suitable weak or limiting branch states, the same formulae are used in the integrated or distributional sense obtained by applying the smooth identities to the regularized representatives supplied by the companion compactness and local energy packages, then passing to the branch topology. The exact residuals and the four error functionals in Theorem 4.2 record the terms that must survive this passage to the limit.

3. A Localized Entropy Functional

Throughout this section, fix an admissible branch state $(V, P, a, b) \in \mathfrak{S}$ represented on $Q_R = B_R \times (-R^2, 0)$, after restricting a Type I ancient profile to Q_R if necessary, and fix a terminal parameter $\sigma \in (-R^2, 0)$. For $s \in (-R^2, \sigma)$ set

$$\theta = \theta(s; \sigma) := \sigma - s > 0. \tag{3.1}$$

Fix a nonnegative, nontrivial cutoff $\chi \in C_c^\infty(B_R)$; we keep χ and R fixed throughout the section. All errors involving derivatives of χ are systematically tracked.

3.1. **The normalized backward adjoint weight.**

Definition 3.1 (Normalized backward adjoint weight)**.** A *normalized backward adjoint weight* for (V, P, a, b) on (s_0, σ) is a positive function $\rho \in C^2(B_R \times (s_0, \sigma))$, $s_0 > -R^2$, satisfying

$$-\partial_s \rho - \Delta \rho - \operatorname{div}\big((a(s)y + b(s) - V)\rho\big) + \kappa_{\chi,R}(s)\rho = 0 \qquad \text{in } B_R \times (s_0, \sigma), \tag{3.2}$$

together with the scalar normalization coefficient

$$\kappa_{\chi,R}(s) := \frac{\displaystyle\int_{B_R} \big(\Delta \chi^2 - (a(s)y + b(s) - V) \cdot \nabla \chi^2\big)\rho \, dy}{\displaystyle\int_{B_R} \chi^2 \rho \, dy}, \tag{3.3}$$

taken on intervals where the denominator is positive, together with the normalization

$$\int_{B_R} \chi^2(y)\rho(y, s; \sigma) \, dy = 1 \qquad \text{for every } s \in (s_0, \sigma). \tag{3.4}$$

We write

$$\rho(y, s; \sigma) = (4\pi\theta)^{-3/2} e^{-f(y,s;\sigma)}, \tag{3.5}$$

which uniquely determines $f \in C^2(B_R \times (s_0, \sigma))$ on the support of ρ.

Remark 3.2. The drift $a(s)y + b(s) - V(y, s)$ in (3.2) matches the drift of the renormalized equation (2.1). This is the local analogue of the conjugate heat equation used in the Hamilton–Perelman $\mathcal{W}$-functional. In contrast to that setting, equation (3.2) is solved on a compact spatial domain and the normalization (3.4) is imposed via the cutoff χ. The scalar coefficient $\kappa_{\chi,R}$ is not an additional branch modulation; it is the Lagrange multiplier enforcing the weighted mass constraint. Existence for fixed terminal datum can be obtained by solving the unnormalized adjoint equation and dividing by $\int \chi^2 \rho$, which produces exactly (3.3) on intervals where this mass is nonzero.

Lemma 3.3 (Weighted adjoint mass balance)**.** *For every ρ satisfying (3.2),*

$$\frac{d}{ds}\int_{B_R}\chi^2\rho\,dy = -\int_{B_R}\big(\Delta\chi^2 - (a(s)y + b(s) - V)\cdot\nabla\chi^2\big)\rho\,dy + \kappa_{\chi,R}(s)\int_{B_R}\chi^2\rho\,dy, \tag{3.6}$$

because $\chi \in C_c^\infty(B_R)$. In particular, with $\kappa_{\chi,R}$ defined by (3.3), the normalization (3.4) is preserved once it holds at one time.

Proof. Multiply (3.2) by χ^2, integrate, and integrate by parts. Using $\operatorname{div} V = 0$,

$$\begin{aligned}\frac{d}{ds}\int\chi^2\rho &= \int\chi^2\big(-\Delta\rho - \operatorname{div}((ay+b-V)\rho) + \kappa_{\chi,R}\rho\big)\\ &= -\int\Delta\chi^2\,\rho + \int\nabla\chi^2\cdot(ay+b-V)\rho + \kappa_{\chi,R}\int\chi^2\rho + \text{boundary terms.}\end{aligned}$$

The boundary terms vanish because the support of χ is compactly contained in B_R. This is (3.6). Substituting (3.3) gives zero derivative. □

We now derive the equation satisfied by f. This is the essential piece of preparation for the entropy calculation in Section 4.

Proposition 3.4 (The f-equation)**.** *Let $\rho = (4\pi\theta)^{-3/2}e^{-f}$ satisfy (3.2). Then on $\{\rho > 0\}$,*

$$\partial_s f + \Delta f - |\nabla f|^2 + (a(s)y + b(s) - V)\cdot\nabla f = 3a(s) + \frac{3}{2\theta} - \kappa_{\chi,R}(s). \tag{3.7}$$

Equivalently,

$$-\partial_s f = \Delta f - |\nabla f|^2 + (a(s)y + b(s) - V)\cdot\nabla f - 3a(s) - \frac{3}{2\theta} + \kappa_{\chi,R}(s). \tag{3.8}$$

Proof. By (3.5), $\rho = (4\pi\theta)^{-3/2}e^{-f}$, hence

$$\begin{aligned}\partial_s\rho &= \frac{3}{2}(4\pi\theta)^{-5/2}(4\pi)e^{-f} - (4\pi\theta)^{-3/2}e^{-f}\partial_s f\\ &= \frac{3}{2\theta}\rho - (\partial_s f)\rho,\end{aligned}$$

where we used $\partial_s\theta = -1$, so $\partial_s\big((4\pi\theta)^{-3/2}\big) = (4\pi\theta)^{-3/2}\cdot\frac{3}{2\theta}$. Similarly,

$$\nabla\rho = -(\nabla f)\rho, \qquad \Delta\rho = \big(|\nabla f|^2 - \Delta f\big)\rho.$$

Compute the convective term:

$$\begin{aligned}\operatorname{div}\big((ay+b-V)\rho\big) &= \operatorname{div}(ay+b-V)\,\rho + (ay+b-V)\cdot\nabla\rho\\ &= 3a(s)\,\rho - (ay+b-V)\cdot(\nabla f)\rho,\end{aligned}$$

using $\operatorname{div}(ay) = 3a$, $\operatorname{div} b = 0$, and $\operatorname{div} V = 0$. Substituting these into (3.2) and dividing by $\rho > 0$,

$$-\frac{3}{2\theta} + \partial_s f - \big(|\nabla f|^2 - \Delta f\big) - 3a(s) + (ay+b-V)\cdot\nabla f + \kappa_{\chi,R}(s) = 0,$$

which is equivalent to (3.7). □

Remark 3.5. Equation (3.7) is a Hamilton–Jacobi-type equation with viscosity Δf, nonlinear gradient term $-|\nabla f|^2$, and modulated drift $(ay+b-V)\cdot\nabla f$. In the centered self-similar Type I case $a \equiv \frac{1}{2}$, $b \equiv 0$, $\nu = 1$, the right-hand side becomes $\frac{3}{2} + \frac{3}{2\theta} - \kappa_{\chi,R}$. The added scalar $\kappa_{\chi,R}$ is the cutoff-normalization correction; it disappears in the unlocalized whole-space heat-kernel model and is included in the cutoff error in the local theory below.

3.2. **Pressure normalization.** We write

$$\widetilde{P}(y,s) := P(y,s) - (P)_{B_R}(s), \qquad (P)_{B_R}(s) := \frac{1}{|B_R|}\int_{B_R} P(y,s)\,dy \tag{3.9}$$

for the mean-free local pressure. By the pressure package in Theorem 2.2, this normalization differs from the localized Calderón–Zygmund piece P_{loc} only by a function of time. We use $\widetilde{P}$ inside the entropy integrand because the Calderón–Zygmund continuity bound and the renormalized local energy inequality both reference $\widetilde{P}$.

3.3. **The localized entropy functional.**

Definition 3.6 (Localized entropy functional)**.** Let $(V,P,a,b) \in \mathfrak{S}$, let ρ be a normalized backward adjoint weight, and let f be defined by (3.5). The corresponding *localized entropy* at time s relative to terminal time σ is

$$\mathcal{W}_{\chi,R}[V,P,\rho](s;\sigma) := \int_{B_R} \chi^2\Big[\theta|\nabla V|^2 + \theta|V|^3 + \theta|\widetilde{P}|^{3/2} + \theta|\nabla f|^2 + f - 3\Big]\rho\,dy. \tag{3.10}$$

Lemma 3.7 (Parabolic scaling bookkeeping)**.** *Let $z=\lambda y$, $t=\lambda^2 s$, and $\sigma_\lambda = \lambda^{-2}\sigma$. Set*

$$V_\lambda(y,s) = \lambda V(z,t), \qquad P_\lambda(y,s) = \lambda^2 P(z,t), \qquad \rho_\lambda(y,s) = \lambda^3\rho(z,t),$$
$$\chi_\lambda(y) = \chi(z), \qquad f_\lambda(y,s) = f(z,t), \qquad \theta_\lambda(s;\sigma_\lambda) = \lambda^{-2}\theta(t;\sigma).$$

Then $\rho_\lambda\,dy = \rho\,dz$, $\int \chi_\lambda^2\rho_\lambda\,dy = \int\chi^2\rho\,dz$, and

$$\begin{aligned}
\theta_\lambda|\nabla_y f_\lambda|^2 &= \theta|\nabla_z f|^2,\\
\theta_\lambda|V_\lambda|^3 &= \lambda\,\theta|V|^3,\\
\theta_\lambda|P_\lambda|^{3/2} &= \lambda\,\theta|P|^{3/2},\\
\theta_\lambda|\nabla_y V_\lambda|^2 &= \lambda^2\theta|\nabla_z V|^2.
\end{aligned}$$

Thus the adjoint Fisher part and the cutoff normalization are exactly covariant, while the velocity and pressure terms have the expected local coercive weights in the renormalized branch variables. The entropy identity below uses these terms as a controlled local budget; no standalone scale-invariance assertion is used in the proof.

Remark 3.8 (Heat-kernel normalization). In the unlocalized whole-space model $\rho_*(y,s) = (4\pi\theta)^{-3/2}\exp(-|y|^2/(4\theta))$, one has $f = |y|^2/(4\theta)$ and

$$\int_{\mathbb{R}^3} (\theta|\nabla f|^2 + f)\rho_*\,dy = 3.$$

Therefore the constant -3 normalizes the pure adjoint heat-kernel state to zero when $V\equiv 0$ and $P \equiv 0$.

Remark 3.9. The cutoff χ and the radius R are tracked explicitly: we will show in Section 4 that all derivatives of χ appear inside one named error term $\mathcal{E}^{\mathrm{cut}}_{\chi,R}$ and are controlled by the compact-cylinder Caccioppoli estimate imported in Theorem 2.2.

4. Monotonicity up to Controlled Local Error

We now compute $\frac{d}{ds}\mathcal{W}_{\chi,R}$ along the renormalized equation (2.1), the f-equation (3.7), and the local pressure decomposition imported in Theorem 2.2. To make the monotonicity statement auditable, we first record an exact identity with no implicit commutator terms. The companion papers are used only afterward, to dominate the exact residual by the four named error classes.

4.1. **Setup of the calculation.** For convenience write

$$\mathcal{I}(y,s) := \theta|\nabla V|^2 + \theta|V|^3 + \theta|\widetilde{P}|^{3/2} + \theta|\nabla f|^2 + f - 3, \tag{4.1}$$

so that $\mathcal{W}_{\chi,R} = \int \chi^2 \mathcal{I}\rho\, dy$. We use the following localized transport identity with

$$B(y,s) := a(s)y + b(s) - V(y,s), \qquad L := \partial_s - \Delta + B\cdot\nabla,$$

throughout the monotonicity calculation.

Lemma 4.1 (Localized transport identity)**.** *Let ρ be a smooth positive solution of* (3.2)*. For any smooth scalar density F, define*

$$\mathfrak{C}_{\chi,R}[F](s) := \int_{B_R} \left[-F\Delta\chi^2 - 2\nabla\chi^2\cdot\nabla F + FB\cdot\nabla\chi^2\right]\rho\, dy, \tag{4.2}$$

$$\mathfrak{N}_{\chi,R}[F](s) := \int_{B_R} \chi^2\kappa_{\chi,R}(s)F\rho\, dy. \tag{4.3}$$

Then

$$\frac{d}{ds}\int_{B_R}\chi^2 F\rho\, dy = \int_{B_R}\chi^2 LF\,\rho\, dy + \mathfrak{C}_{\chi,R}[F](s) + \mathfrak{N}_{\chi,R}[F](s). \tag{4.4}$$

Proof. Differentiate under the integral and use $\partial_s\rho$ from (3.2):

$$\begin{aligned}\frac{d}{ds}\int\chi^2F\rho &= \int\chi^2(\partial_sF)\rho + \int\chi^2F\big(-\Delta\rho - \operatorname{div}(B\rho) + \kappa_{\chi,R}\rho\big)\\ &= \int\chi^2(\partial_sF)\rho\, dy + \int\big(-\Delta(\chi^2F) + B\cdot\nabla(\chi^2F)\big)\rho\, dy + \int\chi^2\kappa_{\chi,R}F\rho\, dy,\end{aligned}$$

where the boundary integrals vanish by $\chi\in C_c^\infty(B_R)$; expanding derivatives of χ^2 gives (4.4). ☐

Applying Theorem 4.1 to $F = \mathcal{I}$ gives the exact master identity

$$\frac{d}{ds}\mathcal{W}_{\chi,R} = \int_{B_R}\chi^2L\mathcal{I}\,\rho\, dy + \mathfrak{C}_{\chi,R}[\mathcal{I}](s) + \mathfrak{N}_{\chi,R}[\mathcal{I}](s). \tag{4.5}$$

4.2. **Exact residual and dissipation.** The nonnegative part of the intrinsic calculation is the following dissipation:

$$\boxed{\begin{aligned}\mathcal{D}_{\chi,R}(s;\sigma) := \int\chi^2\Big[&2\nu\theta|\nabla^2V|^2 + |\nabla V|^2 + 6\nu\theta|V||\nabla V|^2 + |V|^3\\ &+ |\widetilde{P}|^{3/2} + 2\theta|\nabla^2 f|^2\Big]\rho\, dy.\end{aligned}} \tag{4.6}$$

The boxed expression is manifestly nonnegative. No spectral-gap or Bakry–Émery Poincaré inequality is used in this version of the entropy identity.

We now define the exact intrinsic residual by

$$\mathfrak{R}^{\text{int}}_{\chi,R}(s;\sigma) := \int_{B_R}\chi^2L\mathcal{I}\,\rho\, dy + \mathcal{D}_{\chi,R}(s;\sigma). \tag{4.7}$$

Equivalently, splitting $\mathcal{I} = \mathcal{I}_{\nabla V} + \mathcal{I}_V + \mathcal{I}_P + \mathcal{I}_f + \mathcal{I}_0$ with

$$\mathcal{I}_{\nabla V} := \theta|\nabla V|^2, \quad \mathcal{I}_V := \theta|V|^3, \quad \mathcal{I}_P := \theta|\widetilde{P}|^{3/2}, \quad \mathcal{I}_f := \theta|\nabla f|^2, \quad \mathcal{I}_0 := f - 3,$$

the residual is the sum of the five displayed scalar defects

$$\mathfrak{R}_{\nabla V} := \int \chi^2 L\mathcal{I}_{\nabla V}\rho + 2\nu\theta \int \chi^2 |\nabla^2 V|^2 \rho + \int \chi^2 |\nabla V|^2 \rho, \tag{4.8}$$

$$\mathfrak{R}_{V} := \int \chi^2 L\mathcal{I}_{V}\rho + 6\nu\theta \int \chi^2 |V||\nabla V|^2 \rho + \int \chi^2 |V|^3 \rho, \tag{4.9}$$

$$\mathfrak{R}_{P} := \int \chi^2 L\mathcal{I}_{P}\rho + \int \chi^2 |\widetilde{P}|^{3/2} \rho, \tag{4.10}$$

$$\mathfrak{R}_{f} := \int \chi^2 L\mathcal{I}_{f}\rho + 2\theta \int \chi^2 |\nabla^2 f|^2 \rho, \tag{4.11}$$

$$\mathfrak{R}_{0} := \int \chi^2 L\mathcal{I}_{0}\rho. \tag{4.12}$$

Thus

$$\mathfrak{R}^{\mathrm{int}}_{\chi,R} = \mathfrak{R}_{\nabla V} + \mathfrak{R}_V + \mathfrak{R}_P + \mathfrak{R}_f + \mathfrak{R}_0.$$

Every term not explicitly present in $\mathcal{D}_{\chi,R}$, including pressure couplings, drift/modulation couplings, first-order Fisher terms, and scalar normalization terms, is therefore visible in one of (4.8)–(4.12). Combining (4.5) and (4.7) gives the exact differential identity

$$\frac{d}{ds}\mathcal{W}_{\chi,R} = -\mathcal{D}_{\chi,R}(s;\sigma) + \mathfrak{R}^{\mathrm{int}}_{\chi,R}(s;\sigma) + \mathfrak{C}_{\chi,R}[\mathcal{I}](s) + \mathfrak{N}_{\chi,R}[\mathcal{I}](s). \tag{4.13}$$

This is the identity used below. All terms outside the displayed dissipation are accounted for by the exact residual $\mathfrak{R}^{\mathrm{int}}_{\chi,R}$ and the two explicit localization functionals $\mathfrak{C}_{\chi,R}$ and $\mathfrak{N}_{\chi,R}$.

4.3. **The four-piece error budget.** We now define the four named error contributions as a domination package for the exact residual in (4.13). This is the precise place where the companion papers enter.

Definition 4.2 (Named error terms)**.** On $(s_1, s_2) \subset (-R^2, \sigma)$, a four-piece error decomposition is a quadruple of nonnegative absolutely continuous interval functionals

$$\mathcal{E}^{\mathrm{cut}}_{\chi,R}, \qquad \mathcal{E}^{\mathrm{prs}}_{\chi,R}, \qquad \mathcal{E}^{\mathrm{mod}}_{\chi,R}, \qquad \mathcal{E}^{\mathrm{T1}}_{\chi,R},$$

with $\mathcal{E}^{\mathrm{T1}}_{\chi,R} \equiv 0$ in the Type II case, such that there are nonnegative densities $\mathfrak{e}^{\mathrm{cut}}, \mathfrak{e}^{\mathrm{prs}}, \mathfrak{e}^{\mathrm{mod}}, \mathfrak{e}^{\mathrm{T1}} \in L^1_{\mathrm{loc}}$ with

$$\mathcal{E}^{\bullet}_{\chi,R}(s_1, s_2) = \int_{s_1}^{s_2} \mathfrak{e}^{\bullet}_{\chi,R}(s)\, ds,$$

and

$$\begin{aligned} &\int_{s_1}^{s_2} \left| \mathfrak{R}^{\mathrm{int}}_{\chi,R}(s;\sigma) + \mathfrak{C}_{\chi,R}[\mathcal{I}](s) + \mathfrak{N}_{\chi,R}[\mathcal{I}](s) \right| ds \\ &\qquad \le \mathcal{E}^{\mathrm{cut}}_{\chi,R}(s_1,s_2) + \mathcal{E}^{\mathrm{prs}}_{\chi,R}(s_1,s_2) + \mathcal{E}^{\mathrm{mod}}_{\chi,R}(s_1,s_2) + \mathcal{E}^{\mathrm{T1}}_{\chi,R}(s_1,s_2). \end{aligned} \tag{4.14}$$

The total error is

$$\mathcal{E}_{\chi,R} := \mathcal{E}^{\mathrm{cut}}_{\chi,R} + \mathcal{E}^{\mathrm{prs}}_{\chi,R} + \mathcal{E}^{\mathrm{mod}}_{\chi,R} + \mathcal{E}^{\mathrm{T1}}_{\chi,R}.$$

Remark 4.3. $\mathcal{E}^{\mathrm{cut}}$ measures derivatives falling on χ and the cutoff normalizer $\kappa_{\chi,R}$, i.e. the terms represented by $\mathfrak{C}_{\chi,R}$ and $\mathfrak{N}_{\chi,R}$. $\mathcal{E}^{\mathrm{prs}}$ measures the components of $\mathfrak{R}^{\mathrm{int}}_{\chi,R}$ involving the local pressure decomposition and its harmonic remainder. $\mathcal{E}^{\mathrm{mod}}$ measures the components of $\mathfrak{R}^{\mathrm{int}}_{\chi,R}$ carrying $a(s)$, $b(s)$, and moving-cutoff modulation. $\mathcal{E}^{\mathrm{T1}}$ measures the Type I rescaling defect for centered ancient profiles only. Each term will now be controlled by a named companion theorem.

4.4. **Companion-theorem control of the four error terms.**

Theorem 4.4 (Imported companion domination of the exact residual)**.** *For every admissible represented branch, every cutoff $\chi \in C_c^\infty(B_R)$ with $\chi \equiv 1$ on $B_{R/2}$, and every interval $(s_1, s_2) \Subset (-R^2, \sigma)$, the companion estimates provide a four-piece error decomposition in the sense of Theorem 4.2. More precisely:*

(i) *The cutoff and normalizer terms $\mathfrak{C}_{\chi,R}[\mathcal{I}]$ and $\mathfrak{N}_{\chi,R}[\mathcal{I}]$ are included in $\mathcal{E}^{\mathrm{cut}}_{\chi,R}$ and are controlled by the compact cylinder Caccioppoli, cutoff, and renormalized local-energy estimates* [5, Theorem 5.4 and Proposition 5.1].

(ii) *The pressure components of $\mathfrak{R}^{\mathrm{int}}_{\chi,R}$, including the harmonic-pressure remainder from the local decomposition of $\widetilde{P}$, are included in $\mathcal{E}^{\mathrm{prs}}_{\chi,R}$ and are controlled by the pressure decomposition and pressure-normalization comparison* [5, Theorem 3.26 and Lemma 5.3].

(iii) *The terms in $\mathfrak{R}^{\mathrm{int}}_{\chi,R}$ containing the modulation coefficients a, b and the moving cutoff are included in $\mathcal{E}^{\mathrm{mod}}_{\chi,R}$ and are controlled by the fixed- and moving-cutoff corrected monotonicity formulae* [5, Theorems 9.11 and 9.19].

(iv) *In the Type I rescaling case, the rescaling defect is included in $\mathcal{E}^{\mathrm{T1}}_{\chi,R}$ and is controlled by the velocity no-escape estimate* [4, Theorem 6.5]*; in the Type II case this term is identically zero.*

Together these estimates imply the domination inequality (4.14). This theorem is not an additional PDE claim of the entropy paper; it records the precise imported content from the companion manuscripts needed to turn the exact identity (4.13) into monotonicity.

Lemma 4.5 (Assembly of the exact domination inequality)**.** *Assume the four imported bounds in Theorem 4.4. Then the single domination estimate (4.14) follows with no omitted remainder.*

Proof. The exact scalar integrand in (4.14) is first expanded using the displayed identity $\mathfrak{R}^{\mathrm{int}}_{\chi,R} = \mathfrak{R}_{\nabla V} + \mathfrak{R}_V + \mathfrak{R}_P + \mathfrak{R}_f + \mathfrak{R}_0$, together with the explicit cutoff-normalizer terms $\mathfrak{C}_{\chi,R}[\mathcal{I}]$ and $\mathfrak{N}_{\chi,R}[\mathcal{I}]$. The companion estimates quoted in Theorem 4.4(i)–(iv) are assumed precisely for this expanded list of terms: cutoff and normalizer contributions are assigned to $\mathcal{E}^{\mathrm{cut}}$, pressure-decomposition contributions to $\mathcal{E}^{\mathrm{prs}}$, repaired-gauge and moving-window contributions to $\mathcal{E}^{\mathrm{mod}}$, and the Type I rescaling defect to $\mathcal{E}^{\mathrm{T1}}$. Applying those four estimates term by term and then using the triangle inequality in time gives (4.14). No spatial boundary term remains because the transport identity is tested with $\chi \in C_c^\infty(B_R)$. This lemma concerns the monotonicity identity only; the cutoff derivatives appearing in the separate first-variation identity are tracked separately in $\mathcal{B}_{\chi,R}$ and enter later through the imported zero-gradient bridge. □

4.5. **The localized monotonicity identity.** We can now state the main identity.

Theorem 4.6 (Localized monotonicity identity)**.** *Let $(V, P, a, b) \in \mathfrak{S}$ be an admissible represented branch on Q_R, let $\chi \in C_c^\infty(B_R)$ satisfy $\chi \equiv 1$ on $B_{R/2}$, and let $-R^2 < s_1 < s_2 < \sigma < 0$. If ρ is a normalized backward adjoint weight on a time interval containing $[s_1, s_2]$, then*

$$\mathcal{W}_{\chi,R}(s_1;\sigma) - \mathcal{W}_{\chi,R}(s_2;\sigma) \geq \int_{s_1}^{s_2} \mathcal{D}_{\chi,R}(\tau;\sigma)\, d\tau - \mathcal{E}_{\chi,R}(s_1, s_2;\sigma), \tag{4.15}$$

where $\mathcal{D}_{\chi,R} \geq 0$ is given by (4.6), and $\mathcal{E}_{\chi,R} = \mathcal{E}^{\mathrm{cut}} + \mathcal{E}^{\mathrm{prs}} + \mathcal{E}^{\mathrm{mod}} + \mathcal{E}^{\mathrm{T1}}$ is the four-piece error of Theorem 4.2, supplied by Theorem 4.4.

Proof. Integrating the exact identity (4.13) on (s_1, s_2) gives

$$\mathcal{W}_{\chi,R}(s_2;\sigma) - \mathcal{W}_{\chi,R}(s_1;\sigma) = -\int_{s_1}^{s_2} \mathcal{D}_{\chi,R}(\tau;\sigma)\, d\tau + \int_{s_1}^{s_2} \left[\mathfrak{R}^{\mathrm{int}}_{\chi,R} + \mathfrak{C}_{\chi,R}[\mathcal{I}] + \mathfrak{N}_{\chi,R}[\mathcal{I}]\right] d\tau.$$

By Theorem 4.5, the absolute value of the last integral is bounded by $\mathcal{E}_{\chi,R}(s_1, s_2; \sigma)$. Rearranging gives (4.15). Nonnegativity of $\mathcal{D}_{\chi,R}$ is immediate from (4.6). □

Corollary 4.7 (Existence of one-sided entropy limits)**.** *For every admissible singular branch and every fixed local cutoff χ, on every interval where $\mathcal{W}_{\chi,R}(\cdot;\sigma)$ is finite and the four error terms are absolutely continuous and additive in their upper limit, the improper tail error*

$$\mathcal{E}_{\chi,R}(s,\sigma;\sigma) := \lim_{r\uparrow\sigma} \mathcal{E}_{\chi,R}(s,r;\sigma)$$

exists whenever the total error is finite. On such intervals, the function

$$s \longmapsto \mathcal{W}_{\chi,R}(s;\sigma) + \mathcal{E}_{\chi,R}(s,\sigma;\sigma)$$

is nonincreasing as $s \uparrow \sigma$. Consequently, if the corrected entropy and terminal error remain finite up to σ, then $\mathcal{W}_{\chi,R}(\cdot;\sigma)$ has a finite one-sided limit after accounting for the controlled terminal error.

Proof. Fix $s_1 < s_2 < \sigma$. By additivity,

$$\mathcal{E}_{\chi,R}(s_1,\sigma;\sigma) = \mathcal{E}_{\chi,R}(s_1,s_2;\sigma) + \mathcal{E}_{\chi,R}(s_2,\sigma;\sigma).$$

Adding this identity to Theorem 4.6 gives

$$\begin{aligned}
&\mathcal{W}_{\chi,R}(s_1;\sigma) + \mathcal{E}_{\chi,R}(s_1,\sigma;\sigma) \\
&\qquad \geq \mathcal{W}_{\chi,R}(s_2;\sigma) + \mathcal{E}_{\chi,R}(s_2,\sigma;\sigma) + \int_{s_1}^{s_2} \mathcal{D}_{\chi,R}(\tau;\sigma)\, d\tau.
\end{aligned}$$

Thus the corrected entropy is finite and monotone decreasing as time advances toward σ. Monotone convergence yields the asserted one-sided limit. □

Remark 4.8 (Closed-form check on the model heat kernel)*.* Let $V \equiv 0$, $P \equiv 0$, $a \equiv 0$, $b \equiv 0$, and $\rho_* = (4\pi\theta)^{-3/2} \exp(-|y|^2/(4\theta))$, so $f_* = |y|^2/(4\theta)$ and the f-equation (3.7) reduces to $\partial_s f_* + \Delta f_* - |\nabla f_*|^2 = 3/(2\theta)$, satisfied identically. All velocity and pressure terms vanish, and direct computation gives

$$\theta|\nabla f_*|^2 + f_* - 3 = \frac{|y|^2}{4\theta} + \frac{|y|^2}{4\theta} - 3 = \frac{|y|^2}{2\theta} - 3.$$

The Gaussian moments

$$\int_{\mathbb{R}^3} \frac{|y|^2}{2\theta} (4\pi\theta)^{-3/2} e^{-|y|^2/(4\theta)}\, dy = 3, \qquad \int_{\mathbb{R}^3} \rho_*\, dy = 1$$

yield $\mathcal{W}_{1,\infty}[0,0,\rho_*](s;\sigma) = 3 - 3 = 0$, confirming that the constant -3 is the correct dimensional offset. This is the Perelman $\mathcal{W}$-style normalization, transported to the local compact-cylinder setting.

5. First Variation and Vanishing Entropy Gradient

To recover the terminal singular geometries from the entropy functional quantitatively, we use a constrained first variation of $\mathcal{W}$ as a diagnostic. The variational framework is fixed once and for all in this section: V, P, and f are varied independently, while the constraints $\operatorname{div} V = 0$, $\int P = 0$, and $\int \chi^2\rho = 1$ are imposed on the test directions. We do not vary the Calderón–Zygmund pressure map $P[V]$, nor do we vary the adjoint equation itself. Those PDE relations enter later through the branch equation and the companion row-entry package, not through the first variation.

5.1. **Constrained perturbations.** Let $(V, P, a, b) \in \mathfrak{S}$ be an admissible branch state, fix a cutoff $\chi \in C_c^\infty(B_R)$ and a normalized adjoint weight $\rho = (4\pi\theta)^{-3/2}e^{-f}$. We allow simultaneous perturbations

$$V_\varepsilon = V + \varepsilon\varphi, \qquad P_\varepsilon = P + \varepsilon\psi, \qquad f_\varepsilon = f + \varepsilon\eta, \tag{5.1}$$

with $\varphi \in C_c^\infty(B_R; \mathbb{R}^3)$ divergence-free, $\psi \in C_c^\infty(B_R)$ with $\int_{B_R} \psi = 0$, and $\eta \in C_c^\infty(B_R)$ satisfying

$$\int_{B_R} \chi^2 \eta\rho \, dy = 0, \tag{5.2}$$

which preserves the normalization (3.4) to first order, since $\rho_\varepsilon = (4\pi\theta)^{-3/2}e^{-(f+\varepsilon\eta)}$. We do *not* perturb (a, b, ν); these are fixed parameters of the represented branch.

5.2. **The three Euler–Lagrange residuals: full statement.** Recall the integrand $\mathcal{I} = \theta|\nabla V|^2 + \theta|V|^3 + \theta|\widetilde{P}|^{3/2} + \theta|\nabla f|^2 + f - 3$.

Proposition 5.1 (Exact constrained first variation)**.** *For perturbations (5.1) as above, the first variation of $\mathcal{W}_{\chi,R}$ decomposes as*

$$\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \mathcal{W}_{\chi,R}[V_\varepsilon, P_\varepsilon, \rho_\varepsilon](s;\sigma) = \int_{B_R} \chi^2\rho\Big(\mathfrak{M}^{\mathrm{vel}}\cdot\varphi + \mathfrak{M}^{\mathrm{prs}}\,\psi + \mathfrak{M}^{\mathrm{adj}}\,\eta\Big)\,dy + \mathcal{B}_{\chi,R}[\varphi,\psi,\eta], \tag{5.3}$$

where $\rho_\varepsilon = (4\pi\theta)^{-3/2}e^{-(f+\varepsilon\eta)}$. The cutoff contribution is supported where $\nabla\chi \neq 0$:

$$\mathcal{B}_{\chi,R}[\varphi,\psi,\eta] = \int_{B_R} \big[-2\theta\,((\nabla V)\nabla\chi^2)\cdot\varphi - 2\theta\,\eta\,\nabla\chi^2\cdot\nabla f\big]\rho\,dy, \tag{5.4}$$

and the residual representatives, understood as weighted constrained functionals and pointwise on the set where their displayed weights are nonzero, are

$$\mathfrak{M}^{\mathrm{vel}}(y,s) = -2\theta\rho^{-1}\operatorname{div}(\rho\,\nabla V) + 3\theta|V|V, \tag{5.5}$$

$$\mathfrak{M}^{\mathrm{prs}}(y,s) = \tfrac{3}{2}\theta|\widetilde{P}|^{-1/2}\widetilde{P} - \lambda_{\mathrm{prs}}(s)(\chi^2\rho)^{-1}, \tag{5.6}$$

$$\mathfrak{M}^{\mathrm{adj}}(y,s) = -2\theta\rho^{-1}\operatorname{div}(\rho\,\nabla f) + 1 - \mathcal{I} - \Lambda(s). \tag{5.7}$$

The factor $|\widetilde{P}|^{-1/2}\widetilde{P}$ is interpreted as 0 on $\{\widetilde{P} = 0\}$, the continuous derivative of $|\widetilde{P}|^{3/2}$ at the origin. Here

$$\lambda_{\mathrm{prs}}(s) := \frac{1}{|B_R|}\int_{B_R} \chi^2\rho\,\tfrac{3}{2}\theta|\widetilde{P}|^{-1/2}\widetilde{P}\,dy \tag{5.8}$$

is the canonical representative of the mean-free pressure projection, and $\Lambda(s)$ is the Lagrange multiplier

$$\Lambda(s) := \int_{B_R} \chi^2\rho\,\big[-2\theta\rho^{-1}\operatorname{div}(\rho\nabla f) + 1 - \mathcal{I}\big]\,dy \tag{5.9}$$

for the normalized-adjoint constraint (5.2). With these choices, $\chi^2\rho\,\mathfrak{M}^{\mathrm{prs}}$ has zero Lebesgue mean and $\mathfrak{M}^{\mathrm{adj}}$ has zero $\chi^2\rho$-mean.

Proof. Differentiate (3.10) in ε and evaluate at $\varepsilon = 0$. Since this is the independent-variable framework, P does not vary through V, and ρ varies only through f_ε.

The V-terms give

$$\int \chi^2\rho\,\big(2\theta\nabla V : \nabla\varphi + 3\theta|V|V\cdot\varphi\big)\,dy.$$

Integrating only the interior weight ρ by parts gives

$$2\theta \int \chi^2 \rho \, \nabla V : \nabla \varphi = \int \chi^2 \rho \left[-2\theta \rho^{-1} \operatorname{div}(\rho \nabla V) \right] \cdot \varphi \, dy - 2\theta \int \rho \left((\nabla V) \nabla \chi^2 \right) \cdot \varphi \, dy.$$

Thus the first term in (5.4) is exactly the cutoff correction not included in the interior residual (5.5). Since φ is divergence-free, the interior representative is understood modulo gradients.

The pressure term gives

$$\int \chi^2 \rho \, \tfrac{3}{2} \theta |\widetilde{P}|^{-1/2} \widetilde{P} \, \psi \, dy.$$

Here $\delta \widetilde{P} = \psi$ because the direction ψ has zero Lebesgue mean. The same constraint means that subtracting $\lambda_{\mathrm{prs}} (\chi^2 \rho)^{-1}$ does not change the functional. The value in (5.8) is chosen only to fix the representative by imposing zero Lebesgue mean on $\chi^2 \rho \, \mathfrak{M}^{\mathrm{prs}}$. This is (5.6).

For the adjoint variable, $\partial_\varepsilon \rho_\varepsilon |_{\varepsilon=0} = -\eta \rho$. Hence

$$\delta_f \mathcal{W} = \int \chi^2 \rho \left(2\theta \nabla f \cdot \nabla \eta + (1 - \mathcal{I}) \eta \right) dy.$$

Again integrating only the interior weight ρ by parts gives

$$2\theta \int \chi^2 \rho \, \nabla f \cdot \nabla \eta = \int \chi^2 \rho \left[-2\theta \rho^{-1} \operatorname{div}(\rho \nabla f) \right] \eta \, dy - 2\theta \int \rho \, \eta \, \nabla \chi^2 \cdot \nabla f \, dy.$$

This gives (5.7) together with the second cutoff term in (5.4). Since $\int \chi^2 \rho \, dy = 1$, the choice (5.9) subtracts the weighted mean of the interior adjoint coefficient; subtracting this constant does not change the functional on test functions satisfying $\int \chi^2 \eta \rho = 0$. □

Remark 5.2. The residuals in (5.5)–(5.7) are interior variational residuals of $\mathcal{W}$, not the Navier–Stokes equation itself. All derivatives of the cutoff have been placed in the single explicit functional $\mathcal{B}_{\chi,R}$, so there is no double counting between $\mathfrak{M}^{\mathrm{vel}}, \mathfrak{M}^{\mathrm{adj}}$ and $\mathcal{B}_{\chi,R}$. The pressure equation, the modulated branch equation, and the adjoint equation are imposed separately through $\mathfrak{S}$ and Theorem 4.4. This avoids mixing two different variational problems: independent (V, P, f) variation and reduced variation through $P[V]$ or $\rho[V]$.

5.3. **Zero-gradient branch states.**

Definition 5.3 (Zero-gradient branch state)**.** An admissible branch state $S = (V, P, a, b) \in \mathfrak{S}$ is a *zero-gradient branch state* if there exist a cutoff $\chi \in C_c^\infty(B_R)$, a normalized adjoint weight $\rho = (4\pi\theta)^{-3/2} e^{-f}$, a row index $k \in \{1, \ldots, 5\}$, and admissible row data $\mathcal{D}_k$, such that the constrained first variation vanishes after the row normalization:

$$\mathcal{Z}_k(S, \rho, \chi; \mathcal{D}_k) = 0 \qquad \text{in } \mathcal{T}_{\mathrm{br}}. \tag{5.10}$$

Here

$$\langle A, \phi \rangle_{\chi^2 \rho} := \int_{B_R} \chi^2 \rho \, A \phi \, dy$$

for scalar residuals, with the analogous dot product for vector residuals, and $\mathcal{Z}_k = 0$ means the following simultaneous conditions.

(i) The pressure residual vanishes in the dual of mean-free pressure variations:

$$\langle \mathfrak{M}^{\mathrm{prs}}, \psi \rangle_{\chi^2 \rho} = 0 \qquad \text{for every } \psi \in C_c^\infty(B_R), \quad \int_{B_R} \psi \, dy = 0.$$

(ii) The adjoint residual vanishes in the dual of normalization-preserving adjoint variations, except in row $k = 5$, where the only allowed nonzero part is the tail measure specified by $\mathcal{D}_5$:

$$\langle \mathfrak{M}^{\mathrm{adj}}, \eta \rangle_{\chi^2\rho} = 0 \qquad \text{for every } \eta \in C_c^\infty(B_R), \quad \int_{B_R} \chi^2 \eta \rho \, dy = 0,$$

after discarding the row-5 tail defect outside every compact subset on which the companion topology has retained mass.

(iii) The velocity residual vanishes in the divergence-free dual after the row-specific normalization. For rows $k = 1, 2, 4$, this says that $\mathfrak{M}^{\mathrm{vel}}$ is a gradient on the retained compact core, on each separated core, or on each selected scale window, respectively. For row $k = 3$, there is a constant $\Omega \in \mathfrak{so}(3)$ in $\mathcal{D}_3$ such that

$$\mathfrak{M}^{\mathrm{vel}} - \theta\big(\Omega V - (\Omega y) \cdot \nabla V\big)$$

is a gradient in the divergence-free dual. For row $k = 5$, the velocity residual is a gradient modulo the tail defect measure in $\mathcal{D}_5$.

(iv) The cutoff first-variation functional satisfies

$$\mathcal{B}_{\chi,R} \to 0 \tag{5.11}$$

in $\mathcal{T}_{\mathrm{br}}$, after the same row normalization.

For $k = 1$ and no defect data, (5.10) reduces to the strict vanishing of $\mathfrak{M}^{\mathrm{vel}}, \mathfrak{M}^{\mathrm{prs}}, \mathfrak{M}^{\mathrm{adj}}$ and $\mathcal{B}_{\chi,R}$ as constrained functionals.

Remark 5.4. Because the three residuals come with the constraints $\operatorname{div} \varphi = 0$, $\int \psi = 0$, $\int \chi^2 \eta \rho = 0$ on the admissible perturbations, all equalities above are equalities in constrained duals. Thus gradients are invisible to the velocity variation, constants are invisible to the mean-free pressure variation after the projection (5.8), and constants in the adjoint Euler–Lagrange equation are fixed by the multiplier $\Lambda(s)$. Rows (3)–(5) do not change the Euler–Lagrange residuals; they specify the additional symmetry, window normalization, or tail defect that the companion topology permits before testing the same constrained duals.

5.4. **Quantitative correspondence with the five terminal classes.**

Definition 5.5 (Limiting branch topology)**.** The limiting topology used in Table 1 is the companion branch topology $\mathcal{T}_{\mathrm{br}}$. It consists of local strong convergence of V in L^3_{loc}, local strong convergence of $\widetilde{P}$ in $L^{3/2}_{\mathrm{loc}}$ modulo mean-free normalization, weak convergence of the adjoint measures $\rho \, dy$ on compact subsets, convergence of the modulation parameters (a, b) in the topology supplied by the repaired-gauge representation, and convergence of the four interval error functionals in Theorem 4.2. The Type I branch uses the Seregin extraction topology imported in Theorem 2.1; the Type II branch uses the stability and state-decomposition topology imported in Theorem 2.2.

We now state the imported correspondence used by Theorem 5.9. The table records the residual pattern for each terminal class; the row-entry implication itself is imported from the companion state-decomposition theorems and stated explicitly below. In the scale-cascade row we distinguish the *raw modulation carrier* from the normalized error $\mathcal{E}^{\mathrm{mod}}$ appearing in Theorem 6.1.

Theorem 5.6 (Imported entropy-critical zero-gradient bridge)**.** *In the limiting branch topology supplied by Theorems 2.1 and 2.2, every entropy-critical limit obtained from Theorem 4.6 is a zero-gradient branch state in the sense of Theorem 5.3. This is the zero-gradient bridge supplied by the ordered local state-space decomposition and compactness topology* [5, Theorems 10.17 and 10.18; Proposition 3.35]*, together with the residual closure bridge* [3, Theorems 1.14 and 12.1; Corollary 12.2].

Table 1. Quantitative correspondence between zero-gradient branch states and terminal model classes. Each row records the residuals required to vanish, the row-specific modulation or defect datum, and the imported theorem package that closes the corresponding terminal class. The notation $\mathcal{E}^* \to 0$ means the named normalized error term vanishes in the limiting branch topology.

Class	Vanishing residuals & errors	Active modulation	Closing theorem
(1) Compact single-core	$\mathfrak{M}^{\mathrm{vel}} = \theta\nabla\Pi$, $\mathfrak{M}^{\mathrm{prs}} = 0$ in the mean-free dual, $\mathfrak{M}^{\mathrm{adj}} = 0$ in the normalized-adjoint dual, $\mathcal{E}^{\mathrm{cut}}, \mathcal{E}^{\mathrm{prs}}, \mathcal{E}^{\mathrm{T1}} \to 0$, $\mathcal{E}^{\mathrm{mod}} \to 0$	$a = \frac{1}{2}$, $b = 0$ (Type I) or $a, b \to$ const (Type II)	Theorems 2.1 and 2.3
(2) Multiple-core	$\mathfrak{M}^{\mathrm{vel}}$ is a sum of localized gradients $\sum_j \theta\nabla\Pi_j$ supported near separated cores; $\mathfrak{M}^{\mathrm{prs}} = 0$ in the mean-free dual on each core; $\mathfrak{M}^{\mathrm{adj}} = 0$ in the normalized-adjoint dual; $\mathcal{E}^* \to 0$ on each core	a, b bounded; per-core modulation	Theorem 2.3
(3) Rotational / stationary-hull	$\partial_s V = 0$ modulo rotation; in the residual-dual formulation, $\mathfrak{M}^{\mathrm{vel}} - \theta(\Omega V - (\Omega y) \cdot \nabla V) = \theta\nabla\Pi$ with Ω constant; $\mathfrak{M}^{\mathrm{prs}} = 0$ in the mean-free dual; $\mathfrak{M}^{\mathrm{adj}} = 0$ in the normalized-adjoint dual; $\mathcal{E}^* \to 0$	b remains the translation modulation; $\Omega \in \mathfrak{so}(3)$ is a separate residual symmetry parameter	Theorem 2.4
(4) Scale-cascade	$\mathfrak{M}^{\mathrm{vel}} = \theta\nabla\Pi$ holds on each scale window; $\mathfrak{M}^{\mathrm{prs}} = 0$ in the mean-free dual; $\mathfrak{M}^{\mathrm{adj}} = 0$ in the normalized-adjoint dual; $\mathcal{E}^{\mathrm{cut}}, \mathcal{E}^{\mathrm{prs}}, \mathcal{E}^{\mathrm{T1}} \to 0$; normalized $\mathcal{E}^{\mathrm{mod}} \to 0$ on selected windows	raw modulation carrier J_1, J_6, or J_7 before row-specific normalization	Theorem 2.3
(5) Diffuse / tail-defect	$\mathfrak{M}^{\mathrm{vel}} = \theta\nabla\Pi$ modulo a tail defect measure; $\mathfrak{M}^{\mathrm{prs}} = 0$ in the mean-free dual; $\mathfrak{M}^{\mathrm{adj}} = 0$ modulo a tail concentration in ρ; $\mathcal{E}^* \to 0$ inside compact, defect supported in tail	a, b bounded	Theorems 2.3 and 2.4

Theorem 5.7 (Imported zero-gradient row-entry dictionary)**.** *In the limiting branch topology supplied by Theorems 2.1 and 2.2, every admissible zero-gradient branch state satisfying the finite-budget hypotheses of the companion state space enters one of the five rows of Table 1; after applying the order* (1)–(5)*, the row is unique. The exact companion theorem numbers for the five rows are displayed in Theorem 2.5. More precisely:*

(i) *A zero-gradient state with one retained compact core and vanishing cutoff, pressure, modulation, and Type I rescaling errors satisfies the compact single-core hypotheses closed by Theorems 2.1 and 2.3.*

(ii) *A zero-gradient state whose velocity residual splits into finitely many separated localized gradients satisfies the multibubble or separated-profile hypotheses closed by Theorem 2.3.*
(iii) *A zero-gradient state with a residual skew parameter $\Omega \in \mathfrak{so}(3)$, in the sense that the velocity residual is a gradient modulo $\Omega V - (\Omega y) \cdot \nabla V$, satisfies the rotational or stationary-hull hypotheses closed by Theorem 2.4.*
(iv) *A zero-gradient state whose compact-window residuals vanish but whose raw modulation carrier is carried by the canonical windowwise schedule satisfies the scale-cascade hypotheses closed by Theorem 2.3.*
(v) *A zero-gradient state whose compact residuals vanish only modulo a tail defect satisfies the diffuse or tail-defect hypotheses of Theorems 2.3 and 2.4.*

Remark 5.8. Each item is a direct restatement of the row hypotheses in Table 1 together with the imported theorem package that supplies the corresponding state-space entry. The dictionary records the exact external input needed by the entropy argument and does not replace the companion proofs.

Theorem 5.9 (Imported zero-gradient row recovery)**.** *Let $S_\infty \in \mathfrak{S}$ be an entropy-critical limit. Then S_∞ is a zero-gradient branch state in the sense of Theorem 5.3 and belongs to a unique ordered row of Table 1. Conversely, a state satisfying one of the rows of Table 1 satisfies the zero-gradient condition in $\mathcal{T}_{\mathrm{br}}$, with the row-specific normalization in the scale-cascade case and modulo the displayed tail defect in the diffuse/tail row.*

Proof. Theorem 5.6 gives the zero-gradient conclusion for entropy-critical limits. The unique ordered row assignment is exactly Theorem 5.7. Conversely, fix a row of Table 1 and the corresponding limiting state S. Each row specifies which residuals vanish. For every row, the velocity residual is required to be a gradient (possibly summed over cores in row (2), with a rotational correction in row (3), with row-specific scale normalization in row (4), or modulo a tail defect in row (5)); by Theorem 5.4, this is precisely the divergence-free constraint saying $\mathfrak{M}^{\mathrm{vel}} = 0$ modulo the allowed row defect. Similarly $\mathfrak{M}^{\mathrm{prs}} = 0$ in the mean-free dual is the pressure constraint, and $\mathfrak{M}^{\mathrm{adj}} = 0$ in the normalized-adjoint dual, modulo the displayed tail concentration in row (5), is the normalization-preserving constraint. Hence S satisfies (5.10) in $\mathcal{T}_{\mathrm{br}}$, with the row-specific convention already stated in the theorem. The vanishing of $\mathcal{B}_{\chi,R}$ in the limit is part of the imported zero-gradient bridge and row-entry dictionary: the same cutoff compactness estimates that place $\mathfrak{C}_{\chi,R}[\mathcal{I}]$ and $\mathfrak{N}_{\chi,R}[\mathcal{I}]$ in $\mathcal{E}^{\mathrm{cut}}$ also give the dual convergence of the first-variation cutoff functional (5.4) in $\mathcal{T}_{\mathrm{br}}$. □

Remark 5.10. Theorem 5.9 relies on the imported zero-gradient bridge Theorem 5.6 and the imported row-entry package Theorem 5.7. The entropy argument supplies the exact residuals and the four-piece error pattern; the companion state-decomposition theorems supply the conversion from that pattern to one of the five terminal model classes. This is the intended conditional division of labor.

6. Entropy-Critical Limits and the Ordered Classification

6.1. **Entropy-critical limits.**

Definition 6.1 (Entropy-critical limit)**.** An admissible branch state $S_\infty \in \mathfrak{S}$ is *entropy-critical* if there exist admissible represented branch segments $S^{(n)}(\tau)$ on a common normalized cylinder, times $s_n \uparrow \sigma$, and a fixed cutoff $\chi \in C_c^\infty(B_R)$ such that:
(i) $S^{(n)}(s_n) \to S_\infty$ in the companion compactness topology imported in Theorems 2.1 and 2.2;
(ii) $\mathcal{W}_{\chi,R}[S^{(n)}](s_n; \sigma)$ converges to a finite limit;

(iii) there are window lengths $\delta_n \downarrow 0$, with $s_n - \delta_n > -R^2$, such that the dissipation and error contributions of Theorem 4.6 satisfy

$$\int_{s_n-\delta_n}^{s_n} \mathcal{D}_{\chi,R}^{S^{(n)}}(\tau;\sigma)\,d\tau \to 0, \qquad \mathcal{E}_{\chi,R}^{\bullet,S^{(n)}}\big(s_n-\delta_n, s_n;\sigma\big) \to 0 \tag{6.1}$$

for each $\bullet \in \{\mathrm{cut}, \mathrm{prs}, \mathrm{mod}, \mathrm{T1}\}$, with the convention that $\mathcal{E}^{\mathrm{T1}} \equiv 0$ in the Type II case. The superscript $S^{(n)}$ indicates that the dissipation and error functionals are evaluated along the branch segment $S^{(n)}(\tau)$.

Remark 6.2. Theorem 6.1 requires the four error components to vanish individually, not only in total, and only on shrinking terminal windows. This is the window strength obtained from a finite integrated error budget. In particular, a sequence in the scale-cascade row (4) of the table fails the definition unless $\mathcal{E}^{\mathrm{mod}} \to 0$ on the selected windowwise subsequence supplied by Theorem 2.3.

Lemma 6.3 (Shrinking-window selection)**.** *Let $F_0, F_1, \ldots, F_m \geq 0$ be integrable on (s_*, σ). Then there are $s_n \uparrow \sigma$ and $\delta_n \downarrow 0$, with $s_n - \delta_n > s_*$, such that*

$$\int_{s_n-\delta_n}^{s_n} F_j(\tau)\,d\tau \to 0 \qquad \text{for every } j = 0, \ldots, m.$$

Proof. Choose any $\delta_n \downarrow 0$ and $s_n \uparrow \sigma$ with $(s_n - \delta_n, s_n) \subset (s_*, \sigma)$. Absolute continuity of the Lebesgue integral gives the displayed convergence for each fixed F_j. Since the family is finite, the same sequence works for all $j = 0, \ldots, m$. □

Lemma 6.4 (Uniform shrinking-window selection)**.** *Let $s_n \uparrow \sigma$. For each n, let $F_{n,0}, \ldots, F_{n,m} \geq 0$ be nonnegative densities on (s_*, σ). Assume the finite family is uniformly integrable in terminal neighborhoods:*

$$\lim_{\delta\downarrow 0} \sup_n \sum_{j=0}^{m} \int_{\sigma-\delta}^{\sigma} F_{n,j}(\tau)\,d\tau = 0.$$

Then, after passing to a subsequence, there are $\delta_n \downarrow 0$ with $s_n - \delta_n > s_*$ such that

$$\int_{s_n-\delta_n}^{s_n} F_{n,j}(\tau)\,d\tau \to 0 \qquad \text{for every } j = 0, \ldots, m.$$

Proof. Choose $\varepsilon_\ell \downarrow 0$ so that

$$\sup_n \sum_{j=0}^{m} \int_{\sigma-\varepsilon_\ell}^{\sigma} F_{n,j}(\tau)\,d\tau \leq \ell^{-1}.$$

Passing to a subsequence, arrange $\sigma - s_n < \varepsilon_n/2$. Choose

$$0 < \delta_n < \min\{\varepsilon_n/2,\ 1/n,\ s_n - s_*\}.$$

Then $(s_n - \delta_n, s_n) \subset (\sigma - \varepsilon_n, \sigma)$, so each displayed integral is bounded by the corresponding terminal-neighborhood sum, which tends to zero. □

Proposition 6.5 (Entropy compactification)**.** *Every hypothetical singular branch in the companion state space satisfying the finite corrected entropy budget*

$$\sup_{s\in(s_*,\sigma)} \big|\mathcal{W}_{\chi,R}(s;\sigma)\big| < \infty, \qquad \lim_{s\uparrow\sigma} \mathcal{E}_{\chi,R}(s_*, s;\sigma) < \infty$$

for some $s_* < \sigma$, with the four error components supplied by Theorem 4.4, has an entropy-critical subsequential limit.

Proof. Let

$$E_*(\sigma) := \lim_{s\uparrow\sigma} \mathcal{E}_{\chi,R}(s_*, s; \sigma) < \infty.$$

By additivity, the tail error is

$$\mathcal{E}_{\chi,R}(s, \sigma; \sigma) = E_*(\sigma) - \mathcal{E}_{\chi,R}(s_*, s; \sigma),$$

and tends to 0 as $s \uparrow \sigma$. Hence Theorem 4.7 gives a finite one-sided limit for $\mathcal{W}_{\chi,R}(\cdot; \sigma)$. The compactness components of Theorems 2.1 and 2.2 provide subsequential limits in the corresponding state-space topologies. Telescoping

$$\sum_n \big(\mathcal{W}(s_n; \sigma) - \mathcal{W}(s_{n+1}; \sigma)\big) + \sum_n \mathcal{E}_{\chi,R}(s_n, s_{n+1}; \sigma) \geq \sum_n \int_{s_n}^{s_{n+1}} \mathcal{D}\, ds \tag{6.2}$$

along a sequence $s_n \uparrow \sigma$, together with the finiteness of the entropy limit and the finite integrated error budget, gives finite integrated dissipation and finite integrated error near the terminal time for this fixed singular branch. Applying Theorem 6.3 to the dissipation density and the four error densities gives shrinking terminal windows on which each component vanishes. Passing to the subsequence selected by the companion compactness theorem preserves this vanishing. If the companion extraction is formulated instead as a sequence of already rescaled representatives, the same conclusion follows from Theorem 6.4 provided the companion compactness package supplies the stated uniform terminal integrability on the common cylinder. Without that uniform version one applies the fixed-branch selection before rescaling, which is the case used above. The resulting limit is entropy-critical. □

6.2. **Ordered classification of entropy-critical limits.** The point of the entropy formalism is that nonterminal routing rows in the companion decision tree carry one of the four named error channels. Once all four normalized errors vanish on the selected terminal windows, only the five terminal classes of Table 1 remain. The scale-cascade row is included only after its raw modulation carrier has been converted to the row-specific normalized window. The next theorem strengthens the classification: every entropy-critical limit is assigned to the *first* applicable class in a fixed order, eliminating ambiguity in cases where a state could in principle satisfy more than one row of the table.

Theorem 6.6 (Ordered classification of entropy-critical limits)**.** *Every entropy-critical limit S_∞ belongs to at least one of the following five terminal model classes. Its* ordered class *is the first applicable class in the order below, i.e. row j with rows $(1), \dots, (j-1)$ removed:*

(1) ***Compact single-core class.*** *$\mathfrak{M}^{\mathrm{vel}}$ is a single localized gradient $\theta\nabla\Pi$ supported in a compact subset $K_1 \subset B_R$, with K_1 the support of one nonvanishing core; $\mathcal{E}^* \to 0$ uniformly on every compact set inside B_R; modulation (a, b) is either the centered self-similar pair $(1/2, 0)$ (Type I case) or convergent to a constant pair (Type II case).*
(2) ***Multiple-core class.*** *$\mathfrak{M}^{\mathrm{vel}}$ is a finite sum $\sum_{j=1}^{N} \theta\nabla\Pi_j$ with disjoint supports $K_1, \dots, K_N$, each carrying nontrivial critical L^3 mass; $N \geq 2$; modulation per core is bounded.*
(3) ***Rotational or stationary-hull class.*** *$\partial_s V \equiv 0$ modulo a constant skew-symmetric rotation Ω, i.e. $\mathfrak{M}^{\mathrm{vel}} = \theta\nabla\Pi + \theta(\Omega V - (\Omega y) \cdot \nabla V)$ for some constant $\Omega \in \mathfrak{so}(3)$; the translation modulation b is not identified with Ωy; $\mathcal{E}^* \to 0$.*
(4) ***Scale-cascade class.*** *$\mathfrak{M}^{\mathrm{vel}} = \theta\nabla\Pi$ on each scale window and the modulation channel is resolved by the canonical windowwise schedule imported in Theorem 2.3. Before the row-specific window normalization this is the active carrier component J_1, J_6, or J_7; on the selected entropy-critical windows the normalized $\mathcal{E}^{\mathrm{mod}}$ contribution vanishes.*

(5) ***Diffuse or tail-defect class.*** *$\mathfrak{M}^{\mathrm{vel}} = \theta\nabla\Pi$ modulo a defect measure supported in a tail (logarithmic, Young, or coherent critical) of B_R; the velocity converges weakly inside compacts but loses mass to the tail.*

The ordered classes are therefore mutually exclusive by definition, and each entropy-critical limit is assigned to exactly one ordered class.

Proof. **Exhaustiveness.** By Theorem 6.1, an entropy-critical limit S_∞ is obtained on windows where the dissipation and all four error components vanish in the companion branch topology. The imported zero-gradient row-entry dictionary Theorem 5.7 is formulated precisely for this terminal regime: it converts the vanishing entropy-gradient data into a zero-gradient branch state and then assigns that state to one of the five rows of Table 1. Thus Theorem 5.9 places S_∞ in classes (1)–(5).

Order assignment. Apply the criteria in order:

Class (1) fires if $\mathfrak{M}^{\mathrm{vel}}$ is one localized gradient supported on a single compact subset. This is the most restrictive class, and its hypotheses are checked first.

Class (2) fires if $\mathfrak{M}^{\mathrm{vel}}$ decomposes into multiple disjoint localized gradients. This class subsumes class (1) only in the trivial sense $N = 1$, so the raw class (2) criterion is stated with $N \geq 2$.

Class (3) fires if the residual symmetry data admits a constant skew rotation Ω but class (1)/(2) does not apply. Stationarity modulo rotation is tested only after the compact-core alternatives have been removed by the ordering.

Class (4) fires when the modulation channel is the active carrier in the canonical windowwise schedule before the row-specific normalization. The selected entropy-critical windows have normalized $\mathcal{E}^{\mathrm{mod}} \to 0$, so disjointness is determined by the imported carrier labels, not by a failure of the entropy-critical definition.

Class (5) fires when $\mathcal{E}^* \to 0$ inside compact subsets but the velocity loses mass to a tail defect. The residual trichotomy imported in Theorem 2.4 routes active or affine defects back to classes (1)–(4); class (5) is the genuinely tail-supported remainder.

Mutual exclusion. Define the ordered row $\mathcal{C}_j^{\mathrm{ord}}$ as the raw row $\mathcal{C}_j$ minus $\mathcal{C}_1 \cup \cdots \cup \mathcal{C}_{j-1}$. These ordered rows are pairwise disjoint by construction. Since the row-entry dictionary is exhaustive for entropy-critical limits, every such limit lies in exactly one ordered row. □

Corollary 6.7 (Five-class summary table)**.** *The classification of Theorem 6.6 matches the companion state space as follows.*

Terminal class	*Manifestations in the companion state space*	*Closed in*
(1) Compact single-core	*Direct Type I bounded ancient branches; Type II compact single-core and retained compact-test branches*	*Theorems 2.1 and 2.3*
(2) Multiple-core	*Multibubble, separated-profile, and same-point cascade branches*	*Theorem 2.3*
(3) Rotational/stationary-hull	*Rotational relative equilibria and stationary-hull residual branches*	*Theorem 2.4*
(4) Scale-cascade	*Scale-collapse, scale-rigid, cost-divergence, and carrier-routing branches*	*Theorem 2.3*
(5) Diffuse/tail-defect	*Critical-tail defects, diffuse terminal defects, terminal nonactive branches, and exterior concentration*	*Theorems 2.3 and 2.4*

Proof. This is the manifestation column of Table 1, restricted to the limiting topology after $\mathcal{D} \to 0$ and $\mathcal{E}^* \to 0$, with the row-specific window normalization for class (4). □

7. Reduction to the Companion Exclusion Theorems

The entropy formulation becomes effective only after it is paired with the closure results from the companion papers. The closure hypotheses have already been imported in theorem-numbered form in Theorem 2.5.

Theorem 7.1 (Elimination of the terminal classes). *Under the admissibility, normalization, retained-mass, branch-topology, and finite-budget hypotheses of Theorem 2.5, each of the five terminal model classes in Theorem 6.6 is empty:*

(i) *the compact single-core class is excluded by Theorems 2.1 and 2.3;*
(ii) *the multiple-core class is excluded by Theorem 2.3;*
(iii) *the rotational or stationary-hull class is excluded by Theorem 2.4;*
(iv) *the scale-cascade class is excluded by Theorem 2.3;*
(v) *the diffuse or tail-defect class is excluded by Theorems 2.3 and 2.4.*

Proof. This is Theorem 2.5 applied to the row assigned by Theorem 6.6. The role of the entropy formulation is to show that every entropy-critical limit enters exactly one of the rows to which the imported terminal-class exclusion package applies. □

Combining the assembly results, we obtain the entropy version of the three-paper exclusion program.

Corollary 7.2 (Entropy reformulation of the local exclusion program). *Assume the imported packages Theorems 2.1 to 2.4, the exact error domination Theorem 4.4, the zero-gradient bridge Theorems 5.6 and 5.7, the terminal exclusion package Theorem 2.5, and the finite corrected entropy budget hypothesis of Theorem 6.5. Then every entropy-critical subsequential limit of a hypothetical singular branch is assigned to one of the five ordered terminal classes of Theorem 6.6. By Theorem 7.1, each such terminal class is excluded under the stated hypotheses. Hence no hypothetical singular branch satisfying the imported framework and finite-budget hypotheses remains.*

Proof. Combine Theorem 6.5, Theorem 6.6, and Theorem 7.1. □

Remark 7.3. Theorem 7.2 is an entropy-level summary of the companion papers, not a replacement for them. The local pressure decomposition, carrier routing, terminal profile analysis, and residual compactification remain the substantive PDE content. The entropy formalism shows that these arguments fit together as a finite compactification mechanism for one monotone scalar quantity.

8. Comparison with Hamilton–Perelman Monotonicity

The analogy with Hamilton and Perelman is useful but should not be overstated. In Ricci flow, the $\mathcal{W}$-functional is global and geometric, with the scalar curvature appearing explicitly [12, 20]. In the present Navier–Stokes setting, there is no corresponding global geometric functional available at the level of the companion proofs. The pressure is nonlocal, and the branch changes are genuinely local. For that reason, the functional $\mathcal{W}$ is localized from the outset.

Nevertheless, the two theories share a common pattern.

(1) Both use a backward adjoint weight to measure concentration near a candidate terminal time. In Ricci flow, this is the conjugate heat kernel; here, it is the local backward adjoint weight ρ with modulated drift $ay + b - V$.
(2) Both produce a monotone quantity whose defect is a sum of a nonnegative dissipation and explicit local error terms. In Ricci flow, the dissipation is the squared norm of a gradient soliton operator; here, the dissipation $\mathcal{D}_{\chi,R}$ is given explicitly by (4.6) and the four error terms by Theorem 4.2.

(3) In both settings, a critical point of the monotone quantity represents a terminal model geometry. In Ricci flow, gradient shrinking solitons. Here, by Theorems 5.9 and 6.6, one of five terminal model classes.

The decisive difference is that, in the present paper, the critical classes are not derived from an abstract classification theorem. They are read off from the local state-space decomposition already established in the companion manuscripts, and the row-by-row correspondence in Table 1 makes the matching explicit.

9. Concluding Remarks

The companion papers already provide a complete local routing architecture. The point of the present manuscript is to show that this architecture admits a coherent entropy interpretation in standard PDE language, with the calculation of $\frac{d}{ds}\mathcal{W}$ carried out in full and every error term tagged to a named companion theorem. Once the local enstrophy, mean-free pressure, localized critical-norm velocity budget, and backward adjoint weight are assembled into $\mathcal{W}$, the decision tree may be read as a compactness statement for the entropy flow: every singular branch either dissipates into a previously closed local row or converges to one of five terminal model classes.

This viewpoint suggests two further directions. First, it separates the purely local analytic input from the compactification mechanism, which may be useful in formalization or in alternative singularity analyses. In particular, the four-piece error decomposition makes it possible to identify *which* local estimate is responsible for closing *which* branch of the architecture, a feature that should make the proof amenable to LLM-assisted audit and formal verification. Second, it clarifies the role of the local pressure decomposition: the pressure does not obstruct a monotone quantity once it is measured in the local mean-free norms naturally produced by the companion proofs.

Possible extensions. The localized entropy framework is portable to other critical PDE problems with a similar pressure/gauge nonlocality, provided one has a local state-space stratification on which to define the analogue of $\mathcal{W}$. Candidate settings include energy-supercritical wave equations (with the pressure replaced by a nonlocal phase), harmonic map heat flow with bubbling at the terminal time (with the pressure replaced by the Lagrange multiplier of the constraint), and Yang–Mills heat flow with the gauge as the analogue of the modulation (a, b). In each case, the same five-row classification structure is plausible: compact single-core, multiple-core, symmetry-reduced terminal state, scale cascade, diffuse or tail defect. The role of the present paper is to provide the template; the local PDE estimates would have to be redone for each setting.

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